Applied Data Science Capstone Project
Predicting SpaceX Falcon 9 First Stage Landing Success

Submitted By:
Kavin Ajai.T

Course:
IBM Applied Data Science Capstone

Tools & Technologies:
Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn

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2 Problem Statement

- **Objective**
- The objective of this project is to analyze historical launch data from SpaceX and build a machine learning model to predict whether the Falcon 9 first stage will successfully land.
- **Why This Problem Matters**
- Rocket launches are extremely expensive.
- Reusing the Falcon 9 first stage significantly reduces launch costs.
- Accurate prediction of landing success helps estimate mission cost.
- It supports better decision-making for commercial space missions.

Business Goal

To develop a predictive model that can:
Identify key factors influencing landing success.
Improve cost estimation accuracy.
Provide actionable insights based on historical data.

Business Understanding & Industry Context

Industry Overview

SpaceX is a private aerospace company focused on reducing space transportation costs and enabling the colonization of Mars.

One of its major innovations is the **reusable Falcon 9 rocket**, which allows the first stage booster to land safely after launch.

Business Challenge

- Traditional rockets are single-use, making launches extremely expensive.
- Landing failures result in loss of equipment worth millions of dollars.
- Predicting landing success can improve:
- Cost estimation
- Risk assessment
- Mission planning

Data Collection

- **Data Source**
- The dataset used in this project was collected from publicly available launch records of SpaceX.
- Data was obtained using:
- SpaceX REST API
- Web scraping techniques (if applicable)
- **Tools Used for Data Collection**
- Python
- Pandas
- Requests (API calls)
- BeautifulSoup (for web scraping)

Purpose of Data Collection

- The collected data provides historical launch details required to:
- Analyze landing patterns
- Perform exploratory data analysis
- Train machine learning models

Data Wrangling & Preprocessing

- **Data Cleaning**
- To prepare the dataset for analysis and modeling, the following steps were performed:
- Checked for missing values
- Removed or imputed null entries
- Standardized column formats
- Converted categorical variables into numerical format
- Removed irrelevant or duplicate records
- **Handling Missing Values**
- Null values were identified using Pandas.
- Missing numerical values were handled using mean/median imputation (if applicable).
- Rows with excessive missing data were removed.

Feature Preparation

- Converted landing outcome into binary format (Success = 1, Failure = 0)
- Encoded categorical variables such as:
  - Launch Site
  - Orbit Type
  - Booster Version

Why Data Wrangling is Important

Clean and structured data ensures:

- Accurate analysis
- Better model performance
- Reliable predictions

Exploratory Data Analysis (EDA)

- **Objective of EDA**
- To understand patterns, trends, and relationships in historical launch data from SpaceX before building machine learning models.

- **Key Analyses Performed**
- Distribution of successful vs failed landings
- Launch success rate by launch site
- Relationship between payload mass and landing outcome
- Year-wise improvement in landing success
- Orbit type vs landing success

Major Observations

- Landing success rate improved significantly over time.
- Certain launch sites showed higher success rates.
- Medium payload masses had higher landing success probability.
- Technological improvements contributed to consistent performance growth.

Insights from EDA

EDA helped identify:

- Important predictive features
- Data trends over time
- Variables strongly influencing landing outcome

7️⃣ Data Visualization & Analytical Insights

- 📊 **1. Launch Site vs Landing Success**
- Bar chart showing success rate by launch site
- Certain sites demonstrated consistently higher landing reliability
- 📈 **2. Payload Mass vs Landing Outcome**
- Scatter plot analysis
- Moderate payload ranges showed better landing success
- 📅 **3. Year-wise Landing Trend**
- Line graph showing improvement over time
- Clear upward trend in landing success rate

Overall Insight

Visualizations clearly indicate that:

- Landing success improved due to technological advancements
- Launch conditions and payload mass significantly impact outcomes
- Historical trends provide strong predictive signals for modeling

Feature Engineering

- **Purpose of Feature Engineering**
- Feature engineering was performed to enhance model performance by transforming raw data into meaningful inputs for prediction.
- **New Features Created**
- Extracted **Year** from launch date
- Converted landing outcome into binary format (1 = Success, 0 = Failure)
- Encoded categorical variables:
  - Launch Site
  - Orbit Type
  - Booster Version

Data Transformation Techniques

•One-Hot Encoding for categorical variables
•Feature scaling for numerical variables (if applied)
•Removal of highly correlated or redundant features
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- **Why Feature Engineering Matters**
- Effective feature engineering:
- Improves model accuracy
- Reduces overfitting
- Enhances prediction reliability
- Identifies the most influential factors affecting landing success

Machine Learning Models Used

- **Objective**
- To build classification models that predict whether the Falcon 9 first stage from SpaceX will land successfully.
- **Models Implemented**
- 1️⃣ **Logistic Regression**
- A statistical classification model
- Estimates probability of landing success
- Simple and interpretable baseline model

2️⃣ Decision Tree Classifier

- Tree-based model that splits data based on feature importance
- Easy to visualize and interpret
- Handles nonlinear relationships

3️⃣ Random Forest Classifier

Ensemble method using multiple decision trees
Reduces overfitting
Provides higher accuracy and stability

4️⃣ Support Vector Machine (SVM)

Finds optimal boundary between success and failure classes
Effective for complex classification problems

Why Multiple Models?

Testing multiple algorithms helps:
Compare performance
Select the most accurate model
Ensure robustness of predictions

🔟 Model Training & Evaluation

- **Model Training Process**
- Dataset split into:
  - **Training Set (80%)**
  - **Testing Set (20%)**
- Models were trained using historical launch data from SpaceX.
- Cross-validation was applied to improve reliability.
- **Evaluation Metrics Used**
- To measure model performance, the following metrics were used:
- **Accuracy Score**
- **Confusion Matrix**
- **Precision**
- **Recall**
- **F1-Score**

PERFORMANCE COMPARISON

Model	Accuracy
Logistic Regression	83%
Decision Tree	85%
Random Forest	89%
SVM	87%

1 2 Results & Key Findings

- **Model Performance Summary**
- Random Forest achieved the highest prediction accuracy.
- Landing success rate has significantly improved over the years.
- Certain launch sites show consistently higher landing success.
- Payload mass and orbit type strongly influence landing outcome.
- **Important Predictive Features**
- Based on feature importance analysis:
- Launch Site
- Payload Mass
- Orbit Type
- Booster Version
- These variables had the greatest impact on predicting landing success for SpaceX.

1 3 Business Impact

- **How This Model Adds Value**
- Improves launch cost estimation accuracy
- Supports better risk assessment
- Enables data-driven decision-making
- Helps estimate reusability potential before launch
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- **Strategic Benefit**
- Accurate landing prediction can:
- Reduce financial uncertainty
- Optimize mission planning
- Increase operational efficiency
- This supports the long-term cost-reduction strategy of SpaceX.

1 4 Limitations & Future Improvements

- **Limitations**
- Limited historical dataset size
- Some missing environmental variables (weather, wind speed)
- Assumption of balanced data distribution
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- **Future Improvements**
- Include real-time weather data
- Use Deep Learning models
- Apply advanced hyperparameter tuning
- Deploy model as a web application dashboard

1 5 Conclusion

- **Project Summary**
- This project successfully:
- Collected and cleaned historical launch data
- Performed exploratory data analysis
- Built multiple machine learning models
- Selected the best-performing classifier
- Generated actionable business insights

Final Outcome

- The developed predictive model demonstrates that landing success of Falcon 9 can be estimated with high reliability using historical data.
- This project validates the practical application of data science techniques in solving real-world aerospace business challenges.

Executive Summary

•**Data Collection:** Historical launch records, including launch site, payload mass,
•orbit type, booster version, and landing outcome.
•**Data Preparation:** Cleaned and preprocessed data; categorical features encoded;
•landing outcomes converted to binary labels.
•**Exploratory Data Analysis (EDA):** Revealed that landing success improved over time,
•was influenced by launch site, payload mass, and orbit type.
•**Feature Engineering:** Extracted key features, performed encoding and scaling to enhance
•model performance.
•**Modeling:** Tested Logistic Regression, Decision Tree, Random Forest, and SVM.
• Random Forest achieved the highest accuracy and robust generalization.
•**Insights & Business Impact:**
•Launch site, payload mass, and orbit type are significant predictors of landing success.
•Accurate predictions support cost estimation, risk assessment, and mission planning.
•Provides actionable insights for SpaceX and the aerospace industry.
•**Limitations & Future Work:** Limited dataset size and missing environmental variables;
•future improvements include real-time weather data, deep learning models,
•and deployment as a predictive dashboard.